

Permutations and Derangements

Tomasz Brengos

Committers : Aliaksei Kudzelka

1 Permutations

A *permutation* of n elements is an arrangement (ordering) of those elements. For example, there are 6 permutations of the set $\{a, b, c\}$:

$$(a, b, c), \quad (a, c, b), \quad (b, a, c), \quad (b, c, a), \quad (c, a, b), \quad (c, b, a).$$

Since there are 3 choices for the first element, 2 for the second (once the first is chosen), and 1 for the last, by the multiplicative principle there are $3 \cdot 2 \cdot 1 = 3! = 6$ permutations in total.

Factorials and counting. In general, the number of permutations of n (distinct) elements is given by

$$n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1.$$

Partial permutations (k-permutations). Sometimes we only permute k of the n elements, where $1 \leq k \leq n$ (this notion already appeared in the chapter on counting). The number of ways to do this is the *falling factorial*, denoted $n^{\underline{k}}$, and can be found by thinking:

$$n^{\underline{k}} = n \times (n-1) \times \cdots \times (n-k+1).$$

There are k factors in that product. Using factorial notation, we can write

$$n^{\underline{k}} = \frac{n!}{(n-k)!}.$$

(We avoid the also-common notation $P(n, k)$ for this quantity, since in the chapter on partitions $P(n, k)$ denotes the set of number partitions of n into k parts.)

Relationship to combinations. An alternate derivation uses combinations: first *choose* which k elements from the n will appear (that can be done in $\binom{n}{k}$ ways), then *arrange* those k in order (which can be done in $k!$ ways). Hence,

$$n^{\underline{k}} = \binom{n}{k} k!.$$

Since $\binom{n}{k} = \frac{n!}{(n-k)! k!}$, multiplying by $k!$ yields exactly $\frac{n!}{(n-k)!}$, consistent with the direct counting approach.

2 Derangements

A *derangement* of n elements is a permutation where no element remains in its original position. More precisely, if we think of a permutation as a bijection θ on the set $\{1, 2, \dots, n\}$, then θ is a derangement if and only if

$$\theta(k) \neq k \quad \text{for all } k \in \{1, 2, \dots, n\}.$$

Equivalently, a derangement has no fixed points.

For example, for $n = 3$, the permutations of $\{1, 2, 3\}$ are:

$$(1, 2, 3), \quad (1, 3, 2), \quad (2, 1, 3), \quad (2, 3, 1), \quad (3, 1, 2), \quad (3, 2, 1).$$

Among these, the derangements are $(2, 3, 1)$ and $(3, 1, 2)$; the other permutations fix at least one of the elements.

3 Counting Derangements via Inclusion-Exclusion

Let $D(n)$ denote the number of derangements of n elements. We will use the principle of inclusion-exclusion. Suppose we label the elements as $1, 2, \dots, n$, and define A_i to be the set of permutations that fix the element i (i.e. $\theta(i) = i$). Then any derangement is a permutation that lies in none of the sets A_i (for $1 \leq i \leq n$). We have

$$|A_i| = (n-1)!,$$

since if we fix one position i , then we permute the remaining $n-1$ elements freely. In general, for distinct indices $i_1 < i_2 < \dots < i_k$,

$$|A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}| = (n-k)!.$$

By inclusion-exclusion, the size of the union $A_1 \cup A_2 \cup \dots \cup A_n$ is

$$\sum_{k=1}^n (-1)^{k+1} \binom{n}{k} (n-k)!.$$

Hence the number of permutations that do not lie in this union—i.e. the number of derangements—is

$$D(n) = n! - \binom{n}{1}(n-1)! + \binom{n}{2}(n-2)! - \dots + (-1)^n \binom{n}{n}(n-n)!.$$

Thus, a concise closed-form for the number of derangements is

$$D(n) = \sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

Note on the series for e^{-1} :

In Calculus, one learns that the exponential function has a power series expansion

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

Setting $x = -1$ gives

$$e^{-1} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}.$$

Hence,

$$\sum_{k=0}^n \frac{(-1)^k}{k!} \xrightarrow{n \rightarrow \infty} e^{-1}.$$

If you have not taken (or do not recall) a full course in Calculus, think of this as a special case of a well-known infinite series expansion for the exponential function.

Since the partial sums $\sum_{k=0}^n \frac{(-1)^k}{k!}$ converge to e^{-1} as $n \rightarrow \infty$, we conclude that

$$\lim_{n \rightarrow \infty} \frac{D(n)}{n!} = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k}{k!} = e^{-1}.$$

Numerically, this means that for large n , about $1/e \approx 36.8\%$ of all permutations of $\{1, \dots, n\}$ are derangements (i.e. have no fixed points).

4 A Recurrence Relation

We can also show that $D(n)$ satisfies the recurrence

$$D(n) = (n-1)(D(n-1) + D(n-2)), \quad \text{with } D(1) = 0, D(2) = 1.$$

One way to see this: consider where 1 goes in a derangement of $\{1, 2, \dots, n\}$. It can go to any of $n-1$ positions. If 1 goes to position j , then either (i) the element j goes to position 1 (a swap), which reduces the problem to deranging the remaining $n-2$ elements, or (ii) the element j does *not* go to position 1. In case (ii) we must place the $n-1$ elements $\{2, \dots, n\}$ into the $n-1$ positions $\{1, \dots, n\} \setminus \{j\}$, where each element $i \neq j$ is forbidden from its own position i , and the element j is forbidden from position 1. Relabeling position 1 as “position j ” makes every element forbidden from exactly the position carrying its own label, so these placements are precisely the derangements of $n-1$ elements, of which there are $D(n-1)$. Combining the two cases for each of the $n-1$ choices of j yields the recurrence.